Recap: Algebraic closure of convergent sequences.

$(a_n + b_n) \to a + b$, $(\lambda a_n) \to \lambda a$

$(a_n b_n) \to ab$ $\left(\frac{a_n}{b_n}\right) \to a/b$.

Squeeze Theorem ($a_n \leq b_n \leq c_n$ then b_n converges)

Q: What about convergent sequences that we can't guess the value they converge to e.g.

$$a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}, \text{ with } a_1 = 1.$$

This sequence is always increasing (monotone).

Intuition is that increasing bounded sequences are convergent.

Defn 7.1 (Monotone Sequence)

A sequence (a_n) is <u>monotone increasing</u> if $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$.

A sequence (b_n) is <u>monotone decreasing</u> if $b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$.

If a sequence is both monotone increasing and decreasing we say it is <u>monotone</u>.

Ex: (i) Let (a_n) be the sequence given by $a_n = \sqrt{n}$.

Since,

$$a_{n+1} - a_n = \sqrt{n+1} - \sqrt{n}$$

$$= \sqrt{n+1} - \sqrt{n}\left(\frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}}\right)$$

$$= \frac{n+1-n}{\sqrt{n+1} + \sqrt{n}}$$

$$= \frac{1}{\sqrt{n+1} + \sqrt{n}} \geq 0 \quad \forall n \in \mathbb{N}.$$

Hence, (a_n) is monotone increasing.

(ii) Let (b_n) be a sequence given by $b_n = 1/n$.

Then

$$b_n - b_{n+1} = \frac{1}{n} - \frac{1}{n+1}$$

$$= \frac{(n+1) - n}{n(n+1)}$$

$$= \frac{1}{n(n+1)} > 0, \quad \forall n \in \mathbb{N}$$

Hence (b_n) is monotone decreasing.

Theorem 7.3 : Monotone Convergence Theorem

Every bounded monotone sequence is convergent.

Proof : Let (a_n) be a bounded monotone sequence. Thus, for every $n \in \mathbb{N}$ either $a_n \leq a_{n+1}$ or $a_n \geq a_{n+1}$ and $|a_n| < M$ for some $M \in \mathbb{R}^+$. Define the set

$$A = \{ a_n : n \in \mathbb{N} \}.$$

Since (a_n) is bounded, the set A is bounded from above, and clearly non-empty. By the completeness axiom, $\sup(A)$ must exist. Call $\alpha = \sup(A)$. Similarly, A is bounded from below, the infimum exists. Call $\beta = \inf(A)$. It follows, that for every $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ and $N_2 \in \mathbb{N}$ such that

$$\alpha - \varepsilon < a_{N_1}$$

and

$$a_{N_2} < \beta + \varepsilon.$$

If $a_n \leq a_{n+1}$ for all $n \in \mathbb{N}$ it follows that $a_{N_1} < a_n$ for all $n > N_1$. Hence,

$$\alpha - \varepsilon < a_n \leq \alpha < \alpha + \varepsilon$$

$$\Rightarrow |a_n - \alpha| < \varepsilon$$

for all $n > N_1$. Similarly for sequences satisfying $a_{n+1} \leq a_n$ for all $n \in \mathbb{N}$. See notes. Hence (a_n) is convergent. ☐

Ex: Let (b_n) be sequence given by

$$b_{n+1} = \frac{1}{4 - b_n}$$

and $b_1 = 3$. Looking at terms of the sequence we have

$$b_1 = 3, \quad b_2 = \frac{1}{4-3} = 1, \quad b_3 = \frac{1}{4-1} = \frac{1}{3},$$

$$b_4 = \frac{1}{4 - 1/3} = \frac{1}{11/3} = \frac{3}{11}.$$ Looks monotone decreasing.

Let $0 < b_n \leq 3$ for some $n \in \mathbb{N}$. Then

$$1/4 < \frac{1}{4 - b_n} \leq 1.$$

Clearly $0 < b_1 \leq 3$. Thus by induction $0 < b_n \leq 3$ for all $n \in \mathbb{N}$. Similarly assume $b_{n+1} \leq b_n$ for some $n \in \mathbb{N}$. Then

$$4 - b_n \leq 4 - b_{n+1}$$

$$\Rightarrow \quad \frac{1}{4-b_n} \geq \frac{1}{4-b_{n+1}}$$

$$\Rightarrow \quad b_{n+1} \geq b_{n+2}.$$

Since $b_1 = 3$ and $b_2 = 1$ and $1 \leq 3$, by induction $b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$. That is (b_n) is monotone decreasing. Since $0 < b_n$ for all $n \in \mathbb{N}$, (b_n) is bounded below. By the monotone convergence theorem, it follows that (b_n) is convergent. Let $\lim(b_n) = b$. Hence, $\lim(b_{n+1}) = b$ and thus

$$\lim(b_{n+1}) = \lim\left(\frac{1}{4-b_n}\right)$$

$$b = \frac{1}{4-b}. \qquad \text{Algebraic limit theorem}$$

$$\Rightarrow \quad b^2 - 4b + 1 = 0$$

$$\Rightarrow \quad b = 2 \pm \sqrt{3} .$$

Since $b_n \leq 3$ for all $n \in \mathbb{N}$, thus it must be that

$$b = 2 - \sqrt{3} . \quad \boxed{13} .$$